

EEE 416- Microprocessor and Embedded  
Systems Laboratory  
Jan 2026 level-4 Term-1 Section C  
Final Project Presentation

## DESIGN AND IMPLEMENTATION OF A BIDIRECTIONAL BANGLA BRAILLE TRANSLATION SYSTEM

SUBMITTED BY – GROUP 04



Mushfiqur Rahman  
2106181



Chayan Paul  
2106182



Shamin Yeaser  
2106183



Kayes Sami Malitha  
2106184



BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY  
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

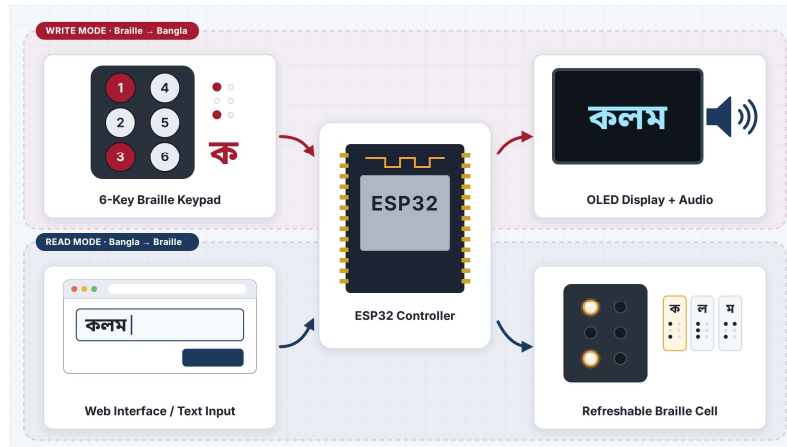
## Outline

- |                                  |                                          |
|----------------------------------|------------------------------------------|
| 1 Summary                        | 6 Reflection on Individual and Team Work |
| 2 Introduction                   | 7 Communication to External Stakeholders |
| 3 Design                         | 8 Project Management and Cost Analysis   |
| 4 Implementation                 | 9 Future Work                            |
| 5 Design Analysis and Evaluation | 10 References                            |



# 1. Summary

**Bidirectional Bangla Braille Communication System** - A device that does both Bangla to Braille and Braille to Bangla translation



## 2. Introduction

### The problem

- Braille is the main reading and writing medium for blind people, but most sighted family members, teachers and classmates cannot read it — a two-way communication gap.
- Commercial refreshable Braille displays are built for other languages and are priced far beyond most Bangladeshi learners [1], [2].
- **Our goal:** a low-cost, single-cell device that works in both directions for Bangla Braille → Bangla text + speech, and Bangla text → tactile Braille.

**US\$ 3,500–15,000**

Typical price of a refreshable Braille display [1]

**US\$ 449**

Cheapest 20-cell display on the market we found (Orbit Reader 20) [2]

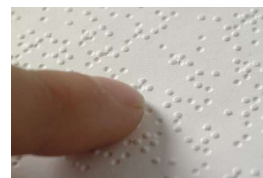


Fig. Embossed Braille text. Source: [3]



Fig. HumanWare Braille display. Source: [4]



## 3.1 Design: Problem Formulation

### 3.1.1 Identification of Scope

- Designing a six-dot physical Braille keyboard.
- Converting Braille patterns into corresponding Bangla characters.
- Supporting Bangla vowels, consonants, kar signs, conjuncts, numbers, punctuation and special Braille sequences.
- Displaying generated Bangla text and providing audio feedback.
- Receiving Bangla text through an ESP32-hosted webpage.
- Converting Bangla text into six-dot Braille patterns and producing physical Braille output using six solenoids.
- Allowing navigation through Braille cells using NEXT/PREVIOUS controls and providing READ/WRITE mode switching.



## 3.1 Design: Problem Formulation

### 3.1.2 Literature Review

| Existing work                                                                      | What it does                                                                         | Gap for our problem                                 |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------|
| Rahman, Rahman & Kaykobad — Bangla Braille Adaptation [5]                          | Defines the Bangla Braille code: letters, kar, conjunct and number rules             | A standard only — no device                         |
| World Braille Usage, 3rd ed. — Bangladesh chart [6]                                | Official Bangla Braille cell for every letter, kar, sign, digit and punctuation mark | Reference chart only — no device                    |
| Kamal et al. — Braille-to-text for Bengali [7]                                     | Scanned embossed Braille → Bangla text (97.25 % accuracy)                            | One direction; needs a scanner; no tactile output   |
| Tabassum et al. — low-cost mechanical Braille display for Bangladeshi students [8] | Refreshable tactile display hardware                                                 | Output only; no Braille typing or text entry        |
| Orbit Reader 20 (commercial) [2]                                                   | 20-cell refreshable display, Bluetooth / USB                                         | US\$ 449; not built around Bangla rules             |
| Arduino refreshable Braille cell project [9]                                       | Single DIY cell driven by a microcontroller                                          | English letters, one direction only                 |
| <b>Our Project</b>                                                                 | Braille ⇄ Bangla in one device, with OLED, audio feedback and web input              | Handles kar, silent ঐ, conjuncts, ঞ / ঙ and numbers |



## 3.1 Design: Problem Formulation

### 3.1.3 Formulation of Problem

There are two main conversion problems:

#### 1. Braille → Bangla

**Workflow :** 6 Braille Keys → Braille Pattern → Bangla Character → OLED + Audio

- Recognize 6-dot Braille patterns
- Handle Bangla characters and combinations
- Provide text and audio output

#### 2. Bangla → Braille

**Workflow :** Bangla Text → Unicode Processing → Braille Pattern → Solenoid Output

- Process Bangla text from webpage
- Convert characters into Braille patterns
- Generate physical Braille using solenoids

**Mode Switching:** Preserve the current text and Braille position when changing modes.



## 3.1 Design: Problem Formulation

### 3.1.4 Analysis

| Decision          | Options considered                       | Chosen and why                                                                          |
|-------------------|------------------------------------------|-----------------------------------------------------------------------------------------|
| Tactile actuator  | Piezo cell, servo, solenoid              | <b>Solenoid:</b> low cost, fast, simple on/off drive                                    |
| Solenoid driver   | Direct GPIO, BJT, logic-level MOSFET     | <b>IRLB8721 MOSFET:</b> full turn-on at 3.3 V, low loss; flyback diode protects it [10] |
| Button inputs     | 9 direct GPIOs, I/O expander             | <b>PCF8575:</b> 9 buttons on the shared 2-wire I <sup>2</sup> C bus, frees GPIOs [10]   |
| Audio feedback    | Cloud text-to-speech, pre-recorded clips | <b>DFPlayer + microSD:</b> instant playback, works offline [10]                         |
| Bangla text input | File upload to flash, live web form      | <b>ESP32 web page:</b> any phone or PC, no app or router needed                         |
| Bangla display    | Hand-made bitmaps, font library          | <b>BanglaText + U8g2:</b> shapes kar and conjuncts automatically [11], [12]             |



## 3.2 Design Methods

### 3.2.1 Hardware components and software

| Component                         | Role in the system                                     |
|-----------------------------------|--------------------------------------------------------|
| ESP32 DevKit (30-pin)             | All logic, Wi-Fi access point and web server           |
| OLED Display                      | Shows Bangla text in both modes                        |
| PCF8575 I/O expander              | Reads 6 dot keys, ENTER/NEXT, BACKSPACE/PREVIOUS, MODE |
| DFPlayer Mini + microSD + speaker | Plays 104 pre-recorded Bangla clips                    |
| 5 V 1A push-pull solenoids × 6    | One refreshable tactile Braille cell                   |
| IRLB8721 MOSFET + 1N5819 × 6      | Logic level switch and flyback protection              |
| LiPo Battery and Buck Converter   | 5 V supply for solenoids and DFPlayer                  |



## 3.2 Design Methods

### 3.2.2 Website Hosting

- ESP32 creates its own Wi-Fi Access Point — no external router needed
- ESP32 works as a Web Server
- Necessary HTML code is written inside the header file **WebPage.h**

Wi-Fi Name  
Braille\_Bangla

Password  
123456789

IP Address  
192.168.4.1

বাংলা ব্রেইল অনুবাদক

বাংলা লিখুন

অনুবাদ করুন

The ESP32-hosted webpage — বাংলা ব্রেইল অনুবাদক



## 3.2 Design Methods

### 3.2.3 Receiving and Cleaning the Input



#### Input Cleaning Rules

- Enter / new-line characters → converted to spaces
- Extra spaces at the start and end → removed

#### Example

Before cleaning

বাংলা  
ব্রেইল



After cleaning

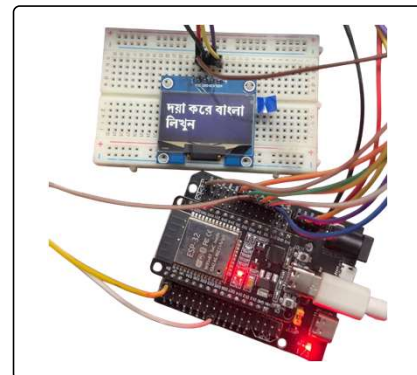
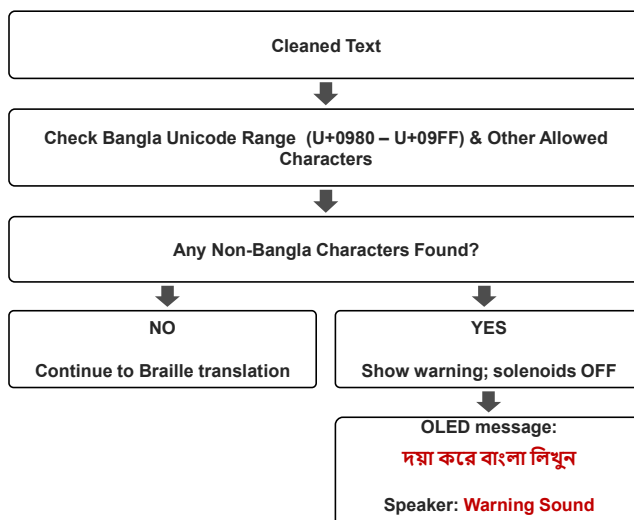
বাংলা ব্রেইল

*handleSend() cleans the text before checking it for Bangla.*



## 3.2 Design Methods

### 3.2.4 Input Text Validation



*showing the Bangla-only warning message*



## 3.2 Design Methods

### 3.2.5 Displaying Bangla Text on the OLED



#### Key idea:

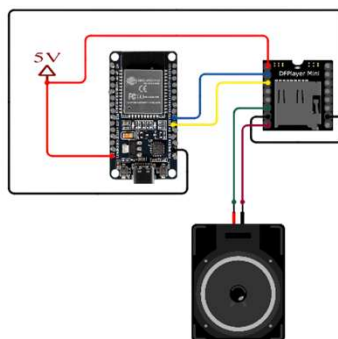
A Bangla text processing library, named **BanglaText** by Mamunul Mazid, is used. The library contains the font header file, which stores the Bangla character glyphs and its pixel mapping.

Source: BanglaText library [11]



## 3.2 Design Methods

### 3.2.6 Audio Output: DFPlayer Mini and Speaker

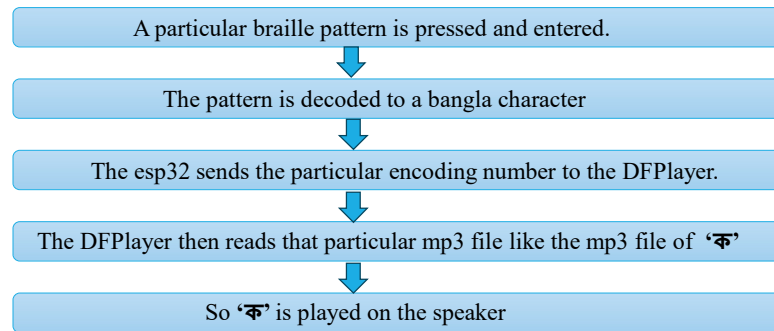


- DFPlayer is used here to read the pre-recorded mp3 files from the microSD card [13]
- 4 ohm, 3 watt speaker is used to play the mp3 files



## 3.2 Design Methods

### 3.2.7 Audio Feedback Workflow



#### Features:

Here we used some other mp3 files for new line, backspace, space, invalid input etc and played that on the speaker. Also there are in total 104 mp3 files. [13]



## 3.2 Design Methods

### 3.2.8 Solenoid Driver Stage (repeated for all six dots)

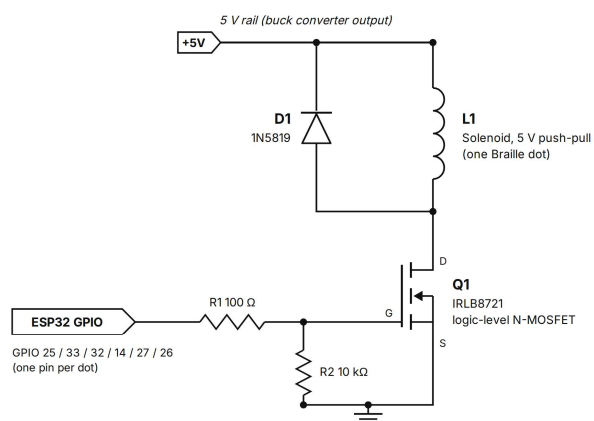


Fig. Solenoid driver circuit

#### MOSFET switch

- IRLB8721 is logic-level: it turns on fully from the 3.3 V GPIO.
- 10 kΩ pull-down keeps the solenoid off while the ESP32 boots; 100 Ω limits gate charging current.

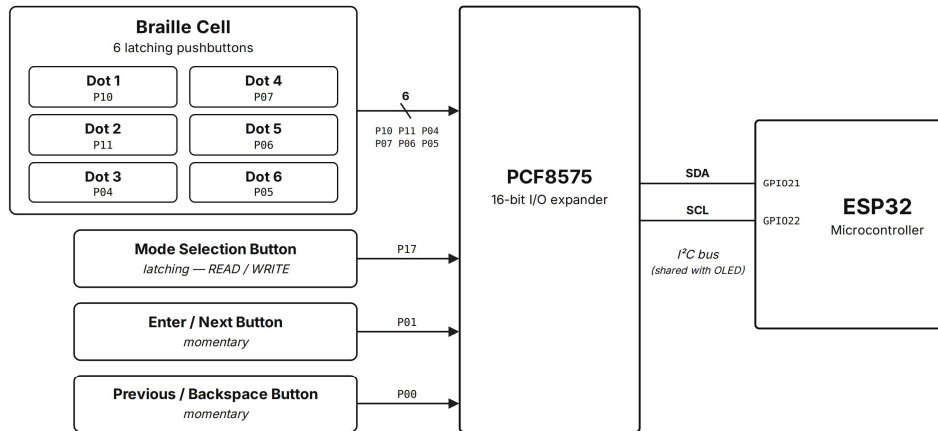
#### Flyback diode

- Switching off a coil gives  $V = L \cdot di/dt$  spikes; the 1N5819 Schottky provides a freewheeling path to handle these spikes.



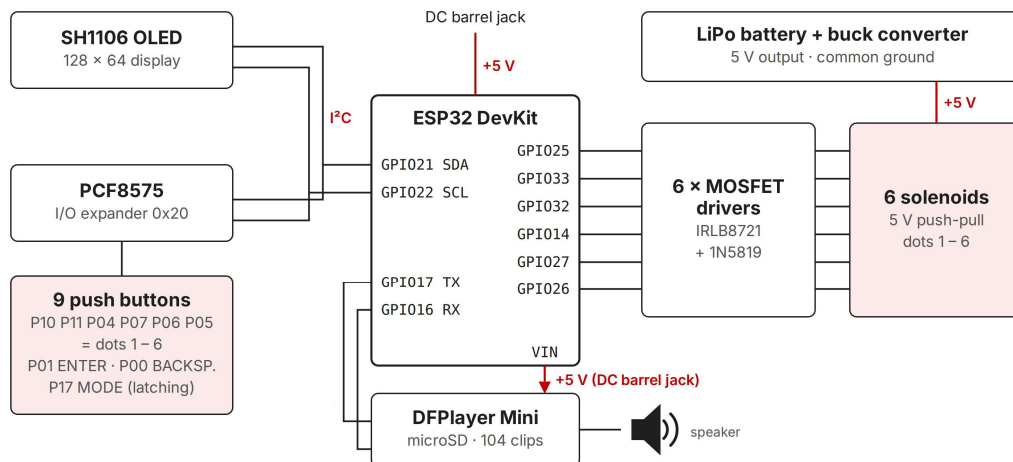
## 3.2 Design Methods

### 3.2.9 Pushbutton Input Subsystem



## 3.2 Design Methods

### 3.2.10 Complete Wiring of the Device



## 3.2 Design Methods

### 3.2.11 Power Budget and Driver Calculations

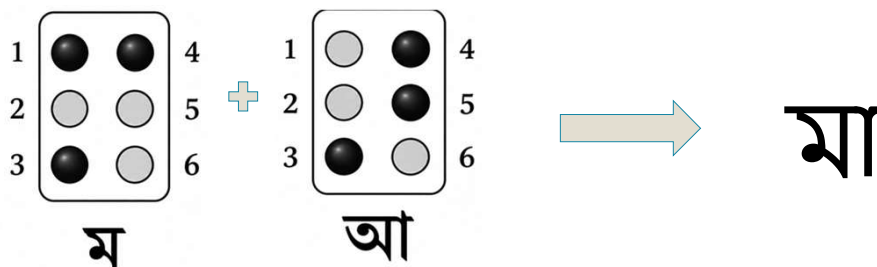
| Load                                            | Current (approx.)     |
|-------------------------------------------------|-----------------------|
| 6 solenoids × 1 A (worst case: all dots raised) | 6.0 A                 |
| ESP32 with Wi-Fi access point (peaks)           | 0.5 A                 |
| DFPlayer Mini + 3 W speaker                     | 0.6 A                 |
| OLED + PCF8575                                  | < 0.05 A              |
| <b>Total</b>                                    | <b>≈ 7.2 A → 36 W</b> |



## 3.2 Design Methods

### 3.2.12 Braille Input – Case 1: Dependent Vowels (কার)

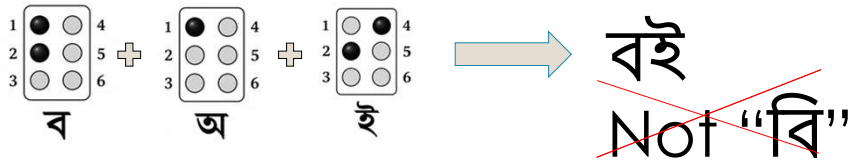
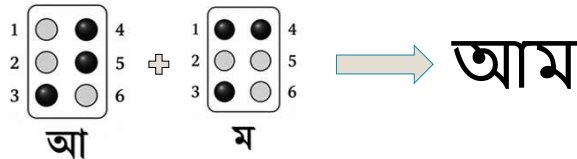
In Braille, independent vowels (ex. আ, ই) and dependent vowels (া, ি) **DO NOT** have a different braille pattern. If a vowel appears directly after a consonant, it is considered as a dependent vowel and is rendered as a কার-attached consonant form. [5], [6]



## 3.2 Design Methods

### 3.2.12 Braille Input – Case 2: Independent Vowels (স্বরবর্ণ)

- If a word starts with a vowel, it is rendered as the standalone vowel.
- If a vowel comes after a closed form consonant ending with “the অ” sound, the vowel is kept as independent and not combined to form a কার. [5], [6]

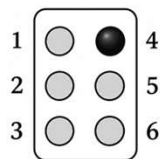


## 3.2 Design Methods

### 3.2.12 Braille Input – Case 3: Conjunct Letters (যুক্তাক্ষর)

In Braille, a যুক্তাক্ষর is NOT implemented using a হসন্ত, rather it uses a special prefix pattern which indicates if the next two or three letters are conjoined. [5], [6]

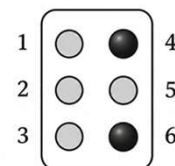
Braille Cell with 4 Raised



Indicates the next two upcoming letters are conjoined



Braille Cell with 4 and 6 Raised



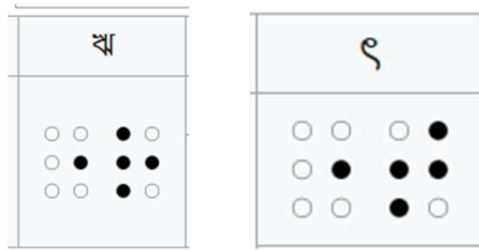
Indicates the next three upcoming letters are conjoined



## 3.2 Design Methods

### 3.2.12 Braille Input – Case 4: Special Characters (ঋ and ঌ)

ঋ and ঌ requires two consecutive Braille cells. But they follow an interesting pattern. If ঋ comes after Dot 5 it becomes ঋ, and if ত comes after Dot 5 it is rendered as ঌ [5], [6]



Dot 5+ ঋ

Dot 5+ ত

We only have one Braille cell, but we can render these two letters by waiting for two consecutive inputs if the first input is a Dot 5 only

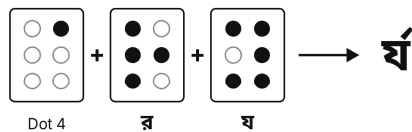


## 3.2 Design Methods

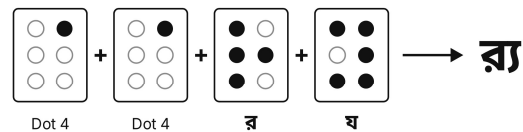
### 3.2.12 Braille Input – Case 5: Conjunct with ঋ (য and ঋ)

Both forms start with the conjunct prefix (Dot 4). Pressing Dot 4 once joins ঋ and য with a hasanta → য (as in কার্য). Pressing Dot 4 twice adds a Zero-Width Joiner (ZWJ) → ঋ (as in র্যাব)

Dot 4 once — normal conjunct  
(ঋ + hasanta + য)



Dot 4 twice — ZWJ conjunct  
(ঋ + ZWJ + hasanta + য)

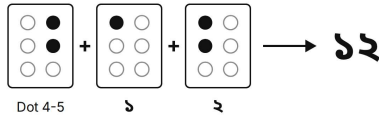


## 3.2 Design Methods

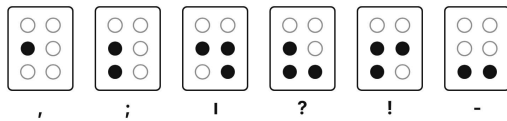
### 3.2.12 Braille Input – Case 6: Numbers and Punctuation

A number starts with the number prefix (Dot 4-5); the following cells are read as digits until a space or a new line. Punctuation marks have their own cells.

Number: Dot 4-5, then digit cells



Punctuation cells

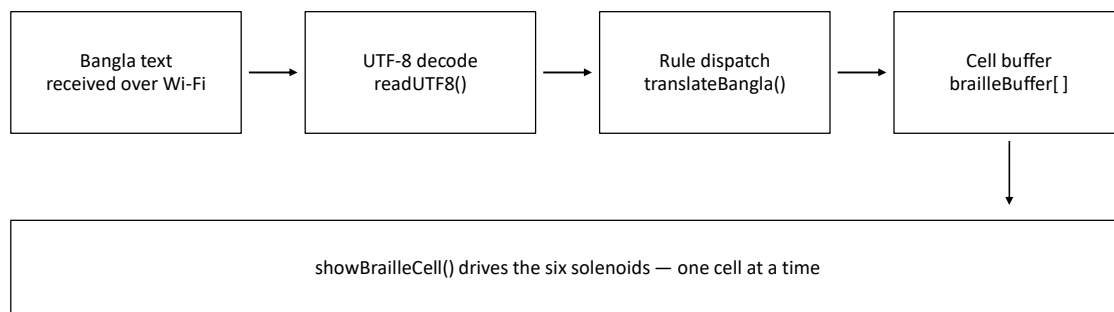


Dots 2-5-6 mean দাঁড়ি (|) in text but a decimal point (.) inside a number: 4-5, 1-4, 2-5-6, 1 → ৩.১. A space ends number mode, so space + 2-5-6 after a number gives ১|



## 3.2 Design Methods

### 3.2.13 Braille Output – Translation Pipeline



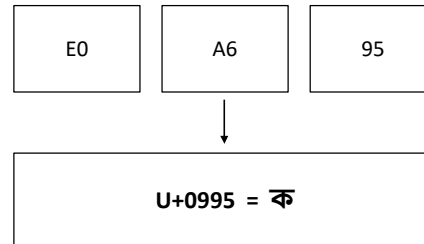
The whole message is translated once. Afterwards, the Next / Previous buttons only move currentCell — nothing is re-translated.



## 3.2 Design Methods

### 3.2.13 Braille Output – Step 1: Decoding the Bangla Text

One Bangla letter = three bytes

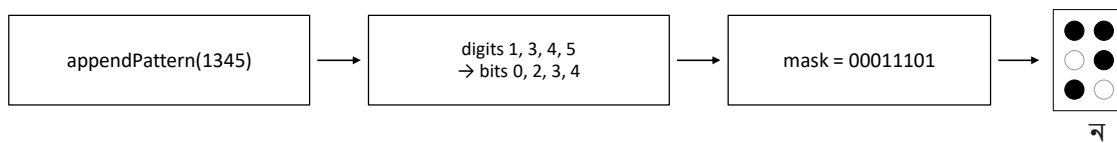


The decoded Bangla character is then matched with the look up table to store the Braille pattern.

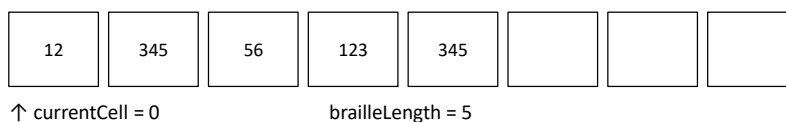


## 3.2 Design Methods

### 3.2.13 Braille Output – Step 2: Storing a Braille Cell



**brailleBuffer[ ]** — one byte per cell, 512 cells maximum



Bit 0 = dot 1 ... bit 5 = dot 6.

Digit order does not matter — the mask is just a set of bits.



## 3.2 Design Methods

### 3.2.13 Braille Output – Step 3: Rule Dispatch Order

1. Space or newline → one blank cell (exits number mode)
2. ঋ and ঌ → a single special sign
3. Any other cluster → prefix cell + each consonant (য → Dot 4 + র + য)
4. Digit → number sign (Dots 4-5) before the first digit, then digit cells
5. "." inside a number → decimal point (Dots 2-5-6)
6. Consonant, kar, punctuation → direct table lookup
7. Independent vowel → ঐ inserted when needed
8. Chandrabindu, anusvara, visarga, ঙ → special signs

Checked top-down.

The first rule that matches wins, consumes its own characters and jumps the index forward.

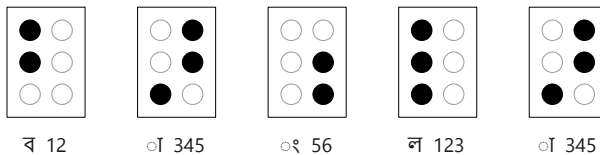
Specific rules must come before general ones — numbers and conjuncts are checked before letters.



## 3.2 Design Methods

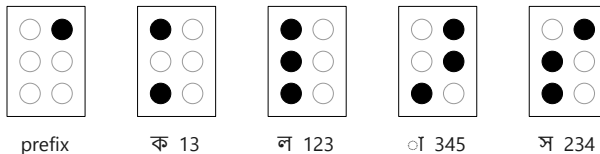
### 3.2.13 Braille Output – Putting It Together

বাংলা → 5 cells



Straight lookup — consonant, kar, consonant. The reader feels one cell at a time and presses Next.

ক্লাস → 5 cells



The dot-4 prefix comes first and warns that the next two letters are joined.



### 3.3 Design: Circuit Diagram

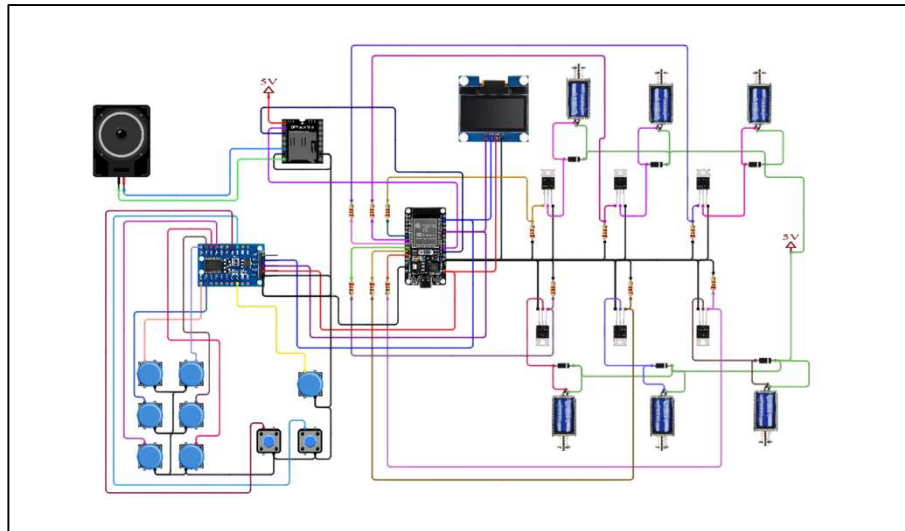


Fig. Complete System Diagram

The Schematic is designed using - [Circuit Designer \[14\]](#)



### 3.4 Design: Circuit Implementation on Veroboard

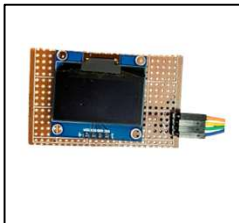


Fig. OLED Connection

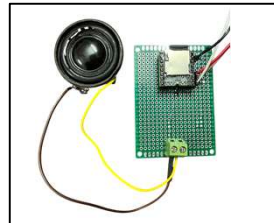


Fig. Speaker Connection

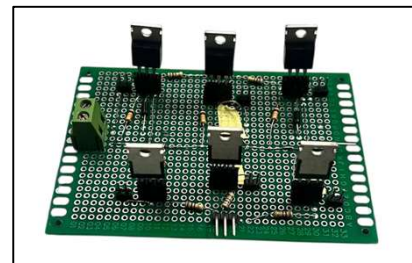
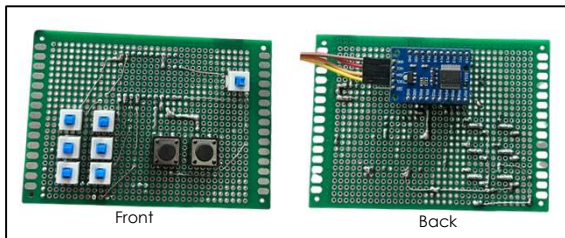


Fig. Solenoid Driving Circuit



Front

Back

Fig. PCF8575 Connection

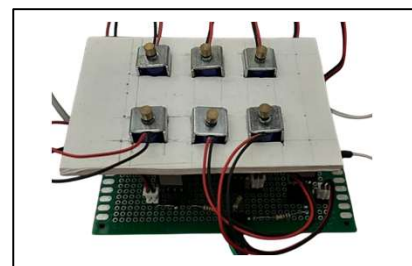
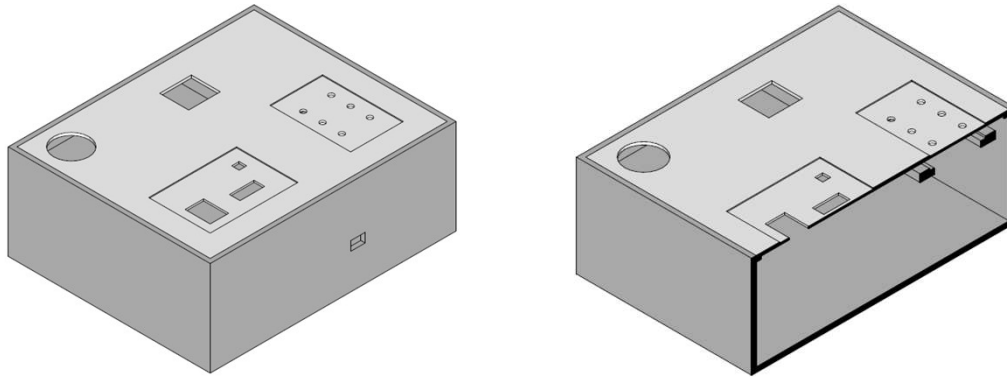


Fig. Solenoid Setup



## 3.5 Design: CAD Model

### 3D Chassis Model



The 3D model design was outsourced under our design guidelines.



## 4. Implementation: Demonstration



Braille to Bangla



Bangla to Braille



## 4.1 Implementation: Photo Gallery



Fig. Startup Message



Fig. Example Entry

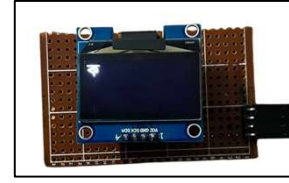


Fig. Bangla Character ক

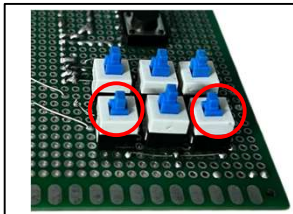


Fig. Corresponding Input Pattern for ক

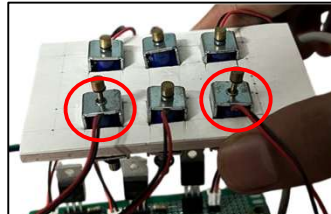


Fig. Corresponding Output Pattern for ক



Fig. Final Setup



## 5. Design Analysis and Evaluation

- 5.1 Novelty
- 5.2 Design Considerations
- 5.3 Investigations
- 5.4 Limitations of Tools
- 5.5 Impact Assessment
- 5.6 Sustainability and Environmental Impact Evaluation
- 5.7 Ethical Issues



## 5.1 Novelty

- **Two-Way Communication:** Bangla ↔ Braille in one device (earlier works are one-way [7], [8], [9])
- **All-in-One System:** Braille input + physical Braille output
- **Multi-Modal Feedback:** Tactile + Visual + Audio
- **Web-Based Input:** Bangla text input without a separate app
- **Bangla Support:** Handles Kar, Hoshonto, digits, punctuation & conjuncts
- **Refreshable Braille:** 6-solenoid physical Braille cell
- **Smart Mode Switching:** READ/WRITE content is preserved



## 5.2 Design Considerations

### Public Health and Safety

- Low-voltage control electronics are used.
- Solenoids require proper driver and protection circuits.
- Electrical connections are insulated from the user.

### Environmental Considerations

- The system uses reusable electronic modules.
- Individual modules can be replaced instead of replacing the entire device.
- Future versions can reduce solenoid power consumption.

### Cultural and Societal Needs

- The system is specifically designed for Bangla language communication for visually impaired people.
- Bangla text is used in the display and webpage.
- Bangla audio feedback improves accessibility.
- The project addresses the communication needs of Bangla speaking Braille users.



## 5.3 Investigations

What we investigated and the design decision each finding led to -

### 5.3.1 Public health and safety

- **Coil heating:** 2 s ON / 1 s OFF drive cuts average coil power to 67 %
- **Voltage spikes:** 1N5822 flyback diode on every coil protects the MOSFET and ESP32
- **Power-up:** 10 k $\Omega$  pull-downs keep all dots down during boot

### 5.3.2 Environment

- **Energy:**  $\approx$  36 W worst case, drawn only while all solenoids are raised
- **Repairability:** modular boards — replace one part, not the device
- **E-waste:** mains supply & rechargeable LiPo battery, no disposable batteries

### 5.3.3 Cultural and societal needs

- **Bangla Braille:** kar, silent ঐ, conjuncts, ঞ / ঙ and numbers follow the standard [5], [6]
- **Bangla display:** BanglaText + Kalpurush font on the OLED [11]
- **Audio:** 104 Bangla clips speak every letter and command [13]
- **Access:** web page on any phone; warnings in Bangla



## 5.4 Limitations of Tools

- **Single Braille Cell:** Long text must be read one character at a time.
- **Limited Memory:** Only a fixed amount of Braille data can be stored.
- **Pre-recorded Audio:** Audio comes from MP3 files on the SD card [13].
- **No Automatic Voice:** Audio for a complete word or sentence cannot be processed.
- **Local Wi-Fi:** The webpage works only through the ESP32 network.
- **Limited Characters:** Some punctuation marks and complex symbols are not implemented for simplicity and to avoid conflicts.
- **High Power:** Solenoids require more current.
- **Slow & Heating:** Solenoids are slower and may heat up with long use.



## 5.5 Impact Assessment

### Societal and Cultural Impact

- Can improve communication between Bangla Braille users and non-Braille users.
- May assist in Bangla Braille education and practice.
- Can be used as a typing device for visually impaired people.

### Health and Safety Impact

- Tactile output can reduce dependence on visual interfaces.
- Solenoid movement, heating and electrical current are controlled safely.

### Legal and Privacy Issues

- User entered text is processed locally by the ESP32 web server.
- The prototype should be presented as an educational assistive system, not as a certified commercial accessibility device.



## 5.6 Sustainability Evaluation

- ESP32 and other modules are reusable and widely available.
- Modular construction allows damaged parts to be replaced separately.
- Local processing reduces dependence on external servers or cloud services.
- The web interface does not require installation of an additional application.



## 5.7 Ethical Issues

- **Standard rule maintained:** The device follows the Bangla Braille code [5], [6] instead of a private code; any deviation forced by the single cell design is documented.
- **Honest claims:** Presented as an educational prototype, not a certified medical or accessibility product. It still needs testing with blind users and Braille teachers.
- **Privacy:** Text is processed only on the ESP32; nothing is sent to the internet or stored.
- **Safety:** Coil current is switched through MOSFETs with flyback diodes, and the ON/OFF cycle limits heating of parts the user touches.
- **Credit:** Open source libraries (U8g2 [12], BanglaText [11], DFRobot DFPlayer [10]), the Bengali-Alphabet audio clips [13] and all papers used are cited in the references.
- **Team ethics:** work was shared openly and every member's contribution is stated in Section 6.



## 6. Reflection on Individual and Team Work

### 6.1 Individual Contribution of Each Member

| Member             | Student ID | Main contribution                                                                                                                                                |
|--------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mushfiqur Rahman   | 2106181    | Bangla text processing(Conjunct, Punctuation, Number, Special character), Text to Braille processing, Circuit soldering, Total system integration, Documentation |
| Chayan Paul        | 2106182    | Braille to audio processing, DFPlayer Bug fix, Solenoid circuit operation, Logistics, System integration, Documentation                                          |
| Shamin Yeaser      | 2106183    | Bangla text processing(Kar, Conjunct), DFPlayer testing, Display setup for Bangla text, System integration, Logistics, Documentation                             |
| Kayes Sami Malitha | 2106184    | Webpage hosting, Circuit soldering, audio file conversion, Logistics, System integration, Documentation                                                          |



## 6. Reflection on Individual and Team Work

### 6.2 Mode of Teamwork and Diversity

- **Divide by subsystem:** The system was built part by part, where the team members worked cooperatively.
- **Test alone, then integrate:** Each part was proven on a breadboard first, then merged into one firmware.
- **Regular reviews:** Progress was checked against the proposed system and the remaining work was reassigned.
- **Diversity:** members brought different strengths — firmware, circuit building, web design and documentation — and decisions were made together.



## 7. Communication to External Stakeholders

| Channel                     | Link / details                                                                                                                                    |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub repository           | <a href="https://github.com/intmushfiq/Bidirectional-Bangla-Braille-Device">https://github.com/intmushfiq/Bidirectional-Bangla-Braille-Device</a> |
| YouTube demonstration video | <a href="https://youtu.be/4rY5ggd-0BE">https://youtu.be/4rY5ggd-0BE</a>                                                                           |



## 8. Project Management and Cost Analysis

### 8.1 Bill of materials

| Item                               | Qty | Unit (₳) | Total (₳)     |
|------------------------------------|-----|----------|---------------|
| ESP32 DevKit (30-pin)              | 1   | 480      | 480           |
| SH1106 OLED 1.3" (PC)              | 1   | 600      | 600           |
| PCF8575 I/O expander module        | 1   | 350      | 350           |
| DFPlayer Mini                      | 1   | 240      | 240           |
| 32 GB microSD                      | 1   | 650      | 650           |
| 4 Ω 3 W speaker                    | 1   | 280      | 280           |
| 5 V push-pull solenoid             | 6   | 250      | 1500          |
| IRLB8721 MOSFET                    | 6   | 150      | 900           |
| 1N5819 diode, resistors            | —   | —        | 50            |
| Push buttons (6 latching, 3 other) | 9   | 10       | 90            |
| LiPo Battery + Buck Converter      | 1   | 2200     | 2200          |
| Veroboard, wires, misc.            | —   | —        | 500           |
| Enclosure (3D printing)            | 1   | 3000     | 3000          |
| <b>Prototype total</b>             |     |          | <b>10,840</b> |

### 8.2 Per-unit cost

**Prototype:** ≈ ₳ 10,800

**Mass-produced:** ≈ ₳ 6,500

Custom PCB instead of modules and breadboards, bulk solenoids, an injection-moulded enclosure and a smaller 5 V supply sized for one cell cut about 35–40 %.



## 9. Future Work

- Add multiple Braille cells for faster reading.
- Design PCB layout for a compact device.
- Add Bangla text-to-speech instead of recorded audio.
- Support more Bangla punctuation marks and symbols.
- Add English or other language processing capability.
- Improve the Braille cell design.
- Reduce power use and heating.
- Test the device with visually impaired users and Braille teachers.



## 10. References

- [1] American Foundation for the Blind, "Refreshable Braille Displays." [Online]. Available: [afb.org/blindness-and-low-vision/using-technology/assistive-technology-products/refreshable-braille](http://afb.org/blindness-and-low-vision/using-technology/assistive-technology-products/refreshable-braille)
- [2] C. Chong, "The Orbit Reader 20: The Most Inexpensive Braille Display," Braille Monitor, NFB, Oct. 2017.
- [3] "ব্রেইল পদ্ধতি," উইকিপিডিয়া (Bangla Wikipedia). [Online]. Available: [https://bn.wikipedia.org/wiki/ব্রেইল\\_পদ্ধতি](https://bn.wikipedia.org/wiki/ব্রেইল_পদ্ধতি) (accessed Sep. 25, 2026).
- [4] Shop4specials Pvt. Ltd., "Humanware Refreshable Braille Display Brailiant BI 40X with Audio," IndiaMART. [Online]. Available: <https://www.indiamart.com/proddetail/humanware-refreshable-braille-display-brailiant-bi-40x-with-audio-27011004962.html> (accessed Sep. 25, 2026).
- [5] M. S. Rahman, M. M. Rahman and M. Kaykobad, "Bangla Braille Adaptation," in Technical Challenges and Design Issues in Bangla Language Processing, IGI Global, 2013, ch. 2, pp. 17–34.
- [6] World Braille Usage, 3rd ed. Watertown, MA, USA: Perkins; Toronto, ON, Canada: International Council on English Braille; Washington, DC, USA: National Library Service for the Blind and Physically Handicapped, Library of Congress, 2013, "Bangladesh," pp. 9–10. ISBN 978-0-8444-9564-4.
- [7] M. Kamal, A. A. Ali, M. A. H. Khan and M. Shoyaib, "Braille to Text Translation for Bengali Language: A Geometric Approach," arXiv:2012.01494, 2020.
- [8] U. W. Tabassum, S. N. Soumma, T. Tasfia and A. S. Gupta, "Design & Fabrication of a Cost-effective, Refreshable Mechanical Braille Display for Visually Impaired Students in Bangladesh," SSRN 4862819.
- [9] Science Buddies, "Build a Refreshable Braille Display with Arduino." [Online]. Available: [sciencebuddies.org](http://sciencebuddies.org)
- [10] Espressif Systems, ESP32 Series Datasheet; Texas Instruments, PCF8575 datasheet; DFRobot, DFPlayer Mini wiki; Infineon, IRLB8721 datasheet.
- [11] M. Mazid, BanglaText — Bangla text rendering library for microcontrollers. [Online]. Available: [github.com/mamunul/BanglaText](https://github.com/mamunul/BanglaText)
- [12] O. Kraus, U8g2 graphics library for monochrome displays. [Online]. Available: [github.com/olikraus/u8g2](https://github.com/olikraus/u8g2)
- [13] M. Zaman (lifeparticle), "Bengali-Alphabet," GitHub repository, MIT License. [Online]. Available: <https://github.com/lifeparticle/Bengali-Alphabet> (accessed Sep. 25, 2026).
- [14] Cirkuit Designer, online circuit design and simulation tool. [Online]. Available: <https://www.cirkuitdesigner.com> (accessed Sep. 25, 2026).



# Thank You

